



Skin Infections

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Learning Objectives:

- Highlight common skin infections and infestations
- Define causative microorganisms and insects of these conditions.
- Identify diagnostic clinical features of each conditions.
- Recognize the social and emotional impact on the sufferer.
- Mention common treatment lines for above mentioned infections and infestations.



Skin infections

- Bacterial
- Fungus/yeasts *→ is a type of fungi*
- Viral including warts
- Infestations *→ parasite
→ protozoa
→ insect*

Viral skin infections:

- Herpes simplex
- Chicken Pox
- Herpes zoster
- Warts and molluscum contagiosum.

Herpes simplex

- Type 1 commonest
- Primary episode
- stomato-gingivitis
- often mild
- Herpes labialis
- *lesion are felt before becoming* **Prodrome: burning** *→ before appearance of lesions*
- **Vesicles and crusting** *→ unless it's recurrent herpes → remission and individual's appearance and treatment differ*
- Self limiting



CHICKEN POX



- Chickenpox is a highly contagious disease caused by the varicella-zoster virus.
- It commonly occurs in children; however, it can also occur in adults. In adults, the symptoms are usually more severe than in children.
- This virus remains in a person's body lifelong, but does not produce symptoms once dormant.



- Symptom is a blistering, itchy skin rash. Outbreaks are more common in winter and early spring.
- Rash characteristics:
 - * Begins as red, raised lesions (papules).
 - Lesions then blister, fill with pus, burst, and scab over (crusting).
- Most prominent on face, scalp, trunk. Less prominent but present on arms, legs

→ usually starting here
- * Itching can be severe.

• INCUBATION PERIOD

The incubation period is about 14 to 21 days after contact with the virus.

** virus may remain dormant in neuronal ganglions and may drop of immunity
it will reappear with a serious presentation (shingles)*



Treatment

Symptomatic treatment for all Children:

- Paracetamol for fever
- Topical calamine can be used for itchiness and it may dry the skin so emollients can be added.
- Antihistamines, can be used orally for itch.
- Parents should be advised to keep children's nails short, cover affected skin with loose light clothing and encourage oral fluids

Herpes zoster (shingles)

- Reactivation of chicken pox virus *(varicella-zoster)*
- Virus in vesicles
- Commoner in elderly compromised
- Dermatomal pattern
- Pain precedes rash *forming shingles*
- 1-3 days later crops of blisters
- Chest neck and forehead commonest sites
- Healing slow in the elderly
- Complications: Post-herpetic neuralgia, blindness *if ophthalmic branch of trigeminal is affected*



Treatment:

to reduce pain systemic
If early, antivirals orally. Topical antiseptics or antibacterials as necessary

Pain relief

Capsaicin

Gabapentin



Common warts



- Human Papilloma Virus
- DNA virus, about > 200 types described till date
- Carcinogenic – anogenital, cervix *could be STD*
- Incubation period: few weeks to about one year
- Transmission: direct or indirect contact (nail biters, shaving, occupational, swimming pool)
- Sexual transmission: genital / perianal wart
- Autoinoculation *like you have it on nail you bite and carriers now you have it in mouth*



Treatment

- No treatment – spontaneous clearing may occur
- Topical Keratolytics – Salicylic acid lotion, wart paints (lactic acid / salicylic Acid)
- Chemical cautery: TCA, *trichloroacetic acid* liquified phenol
- Podophyllin / podophyllotoxin resin – anogenital warts (CI – pregnancy) *→ Contraindicated*
- Cryotherapy (liquid N₂ / CO₂).
- Electrocautery / radiofrequency surgery/ lasers.
- Others: Imiquimod, 1% & 5% 5-FU creams. *→ Topical Chemotherapy*

** genital wart should be removed even in pregnancy since it may enlarge and cause obstructive child birth*

** if external genitalic warts in pregnancy check intrum or birth channel*

Molluscum Contagiosum

بالعربية ١٥/١٠/٢٠٢٥

Symptoms:

- usually asymptomatic
- may have pruritus and/or tenderness

Physical exam:

- immunocompetent patients : single or grouped lesions
- 0.5-1 cm papules with central umbilication, pearly, flesh-colored

Location:

- Trunk, extremities, head, neck & genitals





Treatment:

Treatment is usually not necessary as lesions resolve within 6-9 months.

- **cryotherapy**

well tolerated in adolescents and adults
can be too painful for young children,
especially with multiple lesions

- **topical podophyllotoxin 0.5% cream**
ideal for genital lesions

- **curettage**

Skin Infestations

CUTANEOUS LEISHMANIASIS

*endemic in hot & humid
P. albus*

- has many local synonym such as Tropical sore, Oriental sore, Aleppo sore or Baghdad sore.
- caused by *Leishmania tropica* protozoa, which is endemic in Asia minor, Southwest Asia, the Mediterranean and gulf regions.

- **Modes of Infestations:**

– The **Phlebotomus** sand fly is the vector, transmitting the disease from the reservoirs to human being.

key histology Biological mode of transmission

US history class

– **Direct infection** from infected sores to a traumatized skin may rarely cause the disease

- **Clinical Features:**

- The disease has a very chronic course. The incubation period may take from weeks to two months from the beginning of the sand fly bite .
- Leishmaniasis usually affects children more than other age groups where the face , extremities and the neck are the most common sites involved. *↪ exposed skin*

hard nodule not itchy painful with periphral papules





- Diagnosis: *usually clinical*
 - **Skin biopsy**
 - **Montenegro skin test**
 - uses leishmanial antigen to induce a cell mediated response
 - cannot distinguish between past and present infections
 - Can be false-negative in anergic patients with disseminated infections (*immunocompromised*)
 - **Culture**
 - **Serologic and immunologic tests**



Treatment

- **No treatment** – self-healing lesions
- **Medical:**
 - o Pentavalent antimony (Pentostam), Amphotericin B
 - o Antifungal drugs
 - o +/- Antibiotics for secondary bacterial infection.
- **Surgical:**
 - o Cryosurgery
 - o Excision
 - o Curettage
 - o Lasers (mainly for the scar)

→ best leave scar
→ with early treatment; may prevent scar
→ with proper treatment minimal scar

→ drug of choice



Scabies

→ severe scratching at night
→ whole family affected

- Scabies is caused by the itch mite *Sarcoptes scabiei*.
→ most itchy mite
- The female itch mite tunnels in the topmost layer of the skin and deposits her eggs in burrows.
→ longitudinal papules
→ Pathognomonic feature → only occurs here
→ Diagnostic as it may be seen in other disease
- Young mites (larvae) then hatch in a few days.

- The infestation causes intense itching, probably from an allergic reaction to the mites.
- The infestation spreads from person to person on physical contact, often spreading through an entire household.
- In other cases, mites may be spread on clothing, bedding, and other shared objects, but their survival is brief, and normal laundering destroys them.



Scabies

- The hallmark of scabies is intense itching, which is usually worse at night.
- The burrows of the mites are often visible as very thin lines up to 1/2 inch long, sometimes with a tiny bump at one end.
- Sometimes, only tiny bumps are seen, many of which are scratched open because of the itching. *→ just the papule (dot) → carry of Sarcoptes Scabiei*
- The burrows can be anywhere on the body except the face. *→ if infant face could be affected*

- Common sites are the webs between the fingers and toes, the wrists, ankles, buttocks, nipples, and, in males, the genitals.
- Over time, the burrows may become difficult to see because they are obscured by inflammation induced by scratching.
- People with a weakened immune system may develop severe infestations, which produce large areas of thickened, crusted skin.





Treatment

- cream containing **5% permethrin** which is left on the skin overnight and then washed off. *→ cover whole body from neck to toes*
- For older children and adults, **lindane lotion**. *→ not used for children < 6 years*
- With either drug, a second treatment is required a week later.

** All household contact Must be treated since they might be in incubation period (sexual contact)*

** pillow and stuff disinfected*

- **Ivermectin** taken by mouth in two doses given a week apart also is effective for severe infestations in immunocompromised people

– Even after successful treatment, itching may persist for 2 to 4 weeks because of a continued allergic reaction to the mite bodies. *→ since scabies is in epidermis it takes time to resolve*



Scabies

- The itching can be treated with mild **corticosteroid** cream and **antihistamines** taken by mouth.
- Occasionally, the skin irritation and deep scratches lead to a bacterial infection, which may require **antibiotics** given by mouth.
- Family members and people who have had close physical contact, such as sexual contact, should be treated as well.
- Clothing and bedding used during the preceding few days should be washed in hot water or dry cleaned.

Pediculosis



Pediculosis capitis:

- commonly known as head lice infestation.
- It is caused by the parasitic insect *Pediculus humanus capitis*, which feeds on human blood.
- Symptoms: The primary symptom is **itching**, which may not develop until weeks after the initial infestation.
- Adult head lice are about 2-3 mm long and can be found on the scalp, eyebrows, and eyelashes.

** common in school children can jump from head to head*

** white structure on hair white scale that can be flaked this white stuff (its eggs) are glued like in dust (mid sebaceous dermalis in scalp)*

So if you observe in scalp flick it gently if it starts it is pediculosis if it flies its scale

Treatment



FORMULATION	Therapeutic use	Contraindication
Permethrin based products	Residual activity and increasing drug resistance	Hypersensitivity
Ivermectin Shampoo	Neuromuscular paralysis	None
Benzyl alcohol	Non neurotoxic, non-ovicidal	Hypersensitivity
<i>Lindane 1%</i>	Ovicidal, increasing drug resistance	Premature infants, uncontrolled seizure disorders
Malathion 0.5%	Ovicidal, residual activity, increasing drug resistance	Children less than 6 months, pregnancy
Spinosad	Neuroexcitatory	None



- Treat all members of the family at the same time. Inform the daycare or school.
- The second application of insecticide is required 7–10 days after the first one because the eggs may survive, allowing new louse nymphs to hatch out.
- Physical methods are required to remove the nits, as simple washing is not effective.